Growth, Debt, and Inequality

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Abstract

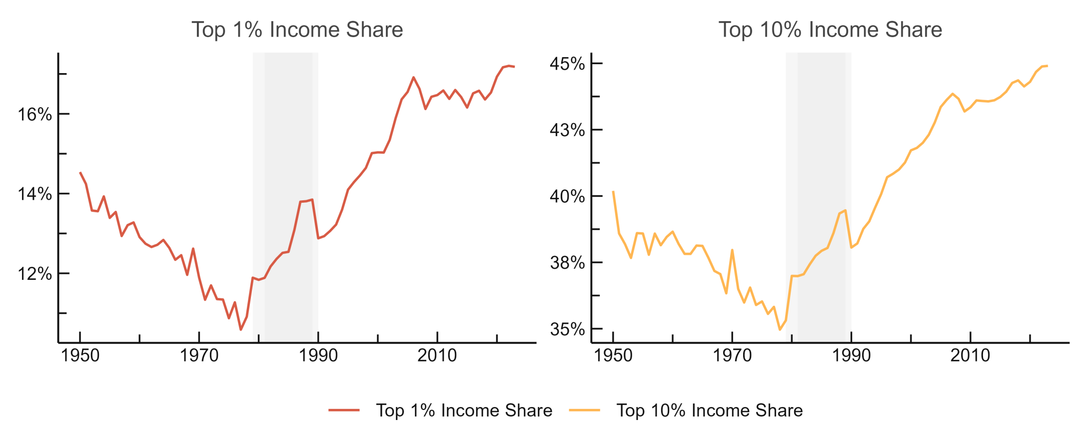
Standard macroeconomic theory assumes that persistent fiscal deficits lead to negative impacts on distributional outcomes. Though there is little empirical evidence to substantiate the assumption of this relationship, it nevertheless perseveres, guiding the study of both fiscal and monetary policy. The most recent global financial crisis has led to an unprecedented increase in public debt across the world, raising serious concerns about its economic impact. In our sample of 109 countries over a period 1978 to 2019, we study the relationship between debt, deficits, and inequality.

1 Introduction

In the aftermath of the Great Recession and the COVID-19 pandemic, public debt levels have soared to unprecedented levels. The extraordinarily high debt levels and seeming fragility of the global economy has brought public and academic debate to reconsider the long-run implications of fiscal spending. Modern arguments can be traced back to Buchanan (1958), Meade (1958), and Modigliani (1961), although the issue goes back as far as Adam Smith. Income and wealth inequality have risen in tandem across much of the world since the early 1980s.

Figure 1 and Figure 2 document both trends using a common cross-country aggregate. Income concentration is measured by the share of pre-tax national income accruing to the top 10% and the top 1% of the distribution, drawn from the World Inequality Database (World Inequality Database, 2024), with the Gini coefficient of disposable income from SWIID 9.91 (Solt, 2020) as a complementary summary measure. Debt is measured at the sectoral level from the IMF Global Debt Database (Mbaye, Moreno-Badia and Chae, 2018), which reports public, household, non-financial corporate, and total private debt as shares of GDP. Each series is aggregated as a GDP-weighted cross-country mean, with weights given by output-side real GDP from the Penn World Tables 11.0 (Feenstra, Inklaar and Timmer, 2015). Because country coverage expands substantially over the sample, unadjusted weighted means confound genuine movements with the entry and exit of countries. We therefore chain the aggregates: for each year, we compute the weighted mean change among countries observed in consecutive years, cumulate these changes over time, and anchor the resulting series to the weighted cross-sectional mean of the final observed year. The aggregates thus track within-country movements while remaining on the scale of the current cross-section.

Figure 1: Long-Run Trends in Debt



Note: Chained GDP-weighted cross-country means, anchored to the final-year cross-section to remove sample-entry effects. Income shares from WID; Gini from SWIID 9.91. Weights are output-side real GDP (PWT 11.0).

Table 1 summarizes these series across three periods: the pre-neoliberal era (1970–1978), the immediate post-neoliberal period (1982–1990), and the most recent decade of the data (2013–2022). The periodization reflects a structural break in advanced-economy fiscal and financial policy that is well documented in the literature.[1] The elections of Margaret Thatcher (1979) and Ronald Reagan (1980) mark the conventional dating of this shift, which encompassed capital account liberalization, credit and banking deregulation, and a reorientation of fiscal priorities, and whose effects propagated to the broader cross-country sample through international capital markets. Rising inequality and rising public debt are both products of this period, and Azzimonti, de Francisco and Quadrini (2014) formalize their joint determination, showing that financial integration and higher income risk together strengthen government incentives to accumulate debt.

The magnitudes in Table 1 are substantial. Public debt nearly doubled between the pre- and post-neoliberal periods, rising from 24% to 47% of GDP, and reached 86% of GDP in the most recent decade. Household debt rose from 40% to 69% of GDP over the full span, and total private debt from 145% to 209%. The distributional series move in tandem: the top 10% income share rose by roughly 3 percentage points between the first two periods and by more than 8 points overall, the top 1% share climbed from 11.3% to 16.9%, and the Gini coefficient rose by 3.5 points. Two patterns stand out. First, the acceleration in both debt and inequality dates to the same narrow window around 1980, which motivates the shaded intervals in Figure 1 and Figure 2. Second, the expansion of debt is broad across sectors rather than confined to the public balance sheet, a fact that motivates our attention throughout the paper to the composition of debt alongside its aggregate level.

Table 1: Debt and Inequality Across Three Periods

	Pre-1979 1970–1978	Post-1981 1982–1990	Recent 2013–2022
Inequality			
Top 10% income share	36.0%	38.9%	44.4%
Top 1% income share	11.3%	13.4%	16.9%
Gini coefficient	32.5	34.3	36.0
Debt (share of GDP)			
Public debt	24.0%	46.6%	85.7%
Household debt	40.0%	49.5%	69.0%
Non-financial corporate	104.9%	114.2%	139.9%
Total private debt	144.9%	163.7%	208.9%

Note: Chained GDP-weighted cross-country means. Aggregates cumulate year-over-year weighted mean changes among countries observed in consecutive years and are anchored to the final-year cross-section, which removes movements attributable to sample entry and exit. Top income shares (pre-tax national income) from WID; Gini coefficient (disposable income) from SWIID 9.91; sectoral debt from the IMF Global Debt Database; weights are output-side real GDP (PWT 11.0).

[1] The early 1980s saw a coordinated acceleration of financial liberalization across OECD economies, documented across multiple dimensions in Abiad, Detragiache and Tressel (2010) and Quinn (1997). The reintegration of international capital markets following the Bretton Woods era is situated in long-run historical perspective in Obstfeld and Taylor (2004). On the fiscal side, Alesina and Perotti (1995) show the wave of spending-based consolidation episodes that characterized OECD fiscal policy in this period. The distributional consequences are documented for the United States in Piketty and Saez (2003) and across a broad set of advanced economies in Atkinson, Piketty and Saez (2011), both of whom identify the early 1980s as the turning point in top income concentration.

Each series is aggregated as a GDP-weighted cross-country mean, with weights given by output-side real GDP from the Penn World Tables 11.0 (Feenstra, Inklaar and Timmer, 2015). The weighting makes the aggregate representative of the world economy rather than of the average country. An unweighted mean assigns the same influence to the smallest economies as to the largest and is therefore dominated by the many small countries in the sample; under GDP weighting, each debt series approximates the ratio of combined debt to combined output, and each distributional series describes income concentration where the bulk of world income is generated. Weighting by output rather than population follows the convention of the cross-country debt literature, in which the object of interest is the fiscal position of the world economy; a population-weighted alternative would characterize the experience of the median person rather than the median unit of output, and would be dominated by China and India. Because country coverage expands substantially over the sample, unadjusted weighted means confound genuine movements with the entry and exit of countries. We therefore chain the aggregates: for each year, we compute the weighted mean change among countries observed in consecutive years, cumulate these changes over time, and anchor the resulting series to the weighted cross-sectional mean of the final observed year. The aggregates thus track within-country movements while remaining on the scale of the current cross-section.

Because the aggregates are GDP-weighted, they are dominated by the high-income economies, and the question arises whether the trends they record extend to the remainder of the sample. Table 2 therefore disaggregate the chained series by World Bank income classification, reporting decade means and linear trends for each group. Top income shares declined through the postwar decades in every group for which the series extend that far back, reached their lowest recorded values in the 1970s and 1980s, and rose thereafter in all four groups. The reversal, however, was not uniform in its incidence. High-income economies paired the largest expansion of public debt, from 40 percent of GDP in the 1980s to 110 percent in the 2020s, with a comparatively moderate distributional trend of roughly one percentage point per decade in the top 10% share. The lower-middle-income group, by contrast, experienced the steepest deterioration on every distributional measure, with the top 10% share rising by 2.5 percentage points per decade since the late 1970s and the Gini coefficient by 2 points per decade, while its public debt accumulated at nearly the same pace as in the low-income group, both near 7 percentage points per decade. Household and corporate debt in the middle-income groups is observed only from the 1990s and expands rapidly thereafter, so that the financial deepening which unfolded across half a century in the high-income economies has been compressed, at much lower levels, into roughly two decades. That the timing of the distributional turn is common to all income groups, while its severity bears no evident relation to the pace of debt accumulation within them, is a first indication of the argument this paper develops: the aggregate level of public debt is a poor predictor of distributional outcomes, which depend instead on the institutional setting and on the composition of borrowing and spending.

The conventional view is that debt can stimulate aggregate demand and output in the short-run, but crowds out capital and reduces output in the long-run (see Mankiw and Elmendorf (1999) for a survey). Standard growth models also predict that higher public debt leads to slower growth (see Saint-Paul (1991) or Aizenman, Kletzer and Pinto (2007)). There are several channels through which high public debt can adversely affect capital accumulation, productivity and growth: higher long-term interest rates and sovereign risk

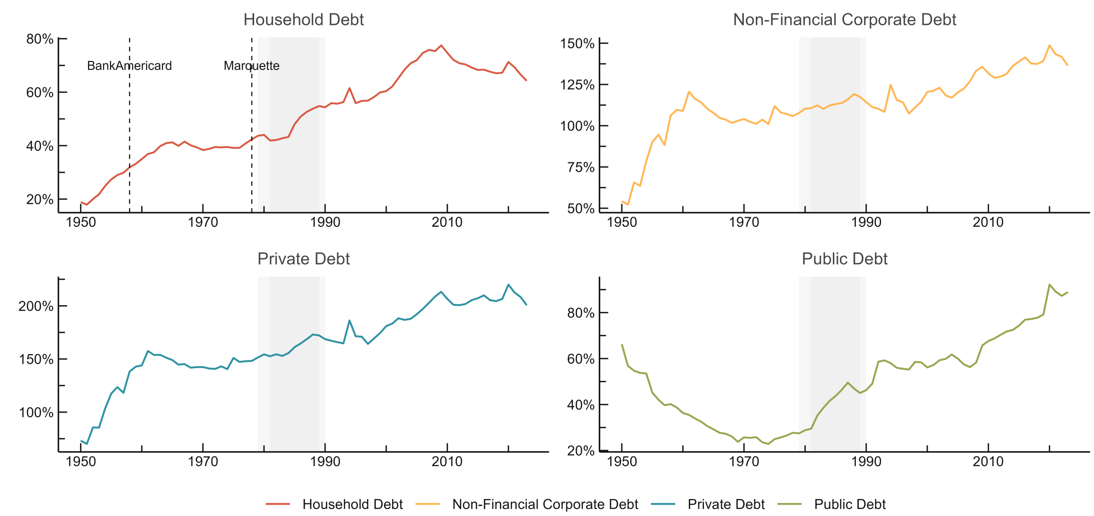
Table 2: Decade Means by Income Group

	1950s	1960s	1970s	1980s	1990s	2000s	2010s	2020s
Top 10% income share								
Low income	45.1%	44.0%	41.9%	40.3%	42.4%	47.1%	53.0%	53.5%
Lower middle income			35.6%	36.3%	39.7%	43.1%	44.0%	45.5%
Upper middle income	48.9%	47.1%	45.3%	42.9%	47.5%	51.3%	50.9%	52.1%
High income	35.0%	34.8%	33.2%	33.9%	36.7%	39.7%	40.7%	41.0%
Top 1% income share								
Low income	17.5%	17.3%	14.5%	13.2%	15.2%	18.6%	19.8%	20.2%
Lower middle income			12.1%	12.6%	14.1%	16.1%	16.2%	17.1%
Upper middle income	18.7%	16.8%	16.4%	16.0%	17.5%	21.0%	21.8%	21.9%
High income	12.8%	11.9%	10.4%	11.2%	12.7%	14.5%	15.0%	15.8%
Gini coefficient								
Low income		42.0	41.8	42.1	43.3	44.9	43.7	41.1
Lower middle income		32.3	33.8	35.1	39.0	42.4	41.9	40.9
Upper middle income		39.5	42.6	43.2	45.1	43.5	40.9	40.1
High income		30.8	29.7	30.0	32.1	33.5	33.8	33.8
Public debt (share of GDP)								
Low income	21.9%	31.5%	34.4%	46.8%	67.7%	65.4%	57.1%	74.6%
Lower middle income	5.1%	12.4%	16.8%	32.9%	39.4%	37.7%	42.5%	67.2%
Upper middle income	31.2%	33.7%	39.8%	58.0%	44.8%	35.1%	36.3%	50.0%
High income	45.1%	30.9%	26.3%	40.3%	58.1%	67.0%	96.3%	110.1%
Household debt (share of GDP)								
Lower middle income					5.2%	19.6%	32.9%	34.8%
Upper middle income					−1.0%	7.0%	17.6%	18.6%
High income	26.7%	39.8%	40.6%	47.9%	60.2%	76.2%	76.2%	73.6%
Non-financial corporate debt (share of GDP)								
Lower middle income					−28.8%	35.0%	97.3%	82.1%
Upper middle income					145.8%	82.2%	110.0%	113.5%
High income	92.3%	118.3%	114.5%	122.7%	118.3%	129.2%	142.3%	149.1%
Total private debt (share of GDP)								
Lower middle income					−23.0%	55.2%	130.8%	117.4%
Upper middle income					144.9%	89.1%	127.5%	132.0%
High income	119.1%	158.0%	155.0%	170.6%	178.5%	205.5%	218.5%	222.7%

Note: Decade means of chained GDP-weighted cross-country means by modal World Bank income classification. The Gini coefficient is reported in index points; all other series as percent of GDP or of national income. The 2020s cover 2020–2023. Sectoral debt series are unavailable for the low-income group, and middle-income sectoral series begin in the 1990s, where they rest on few reporting countries and are sensitive to the chaining anchor. Sources as in Table 1.

spillovers to corporate borrowing costs (Gale and Orszag, 2003; Cecchetti and Kharroubi, 2015; Corsetti et al., 2013), higher future distortionary taxation and lower future public infrastructure spending, higher inflation (Barro, 1995), and greater uncertainty about prospects and policies. High debt levels are also likely to constrain the scope for counter-cyclical fiscal policies, which may result in higher volatility and further lower growth (Woo and Kumar, 2015). In more extreme cases of a debt crisis, by triggering a banking or currency crisis, the adverse effects can be magnified (Reinhart and Rogoff, 2011; Reinhart, Reinhart and Rogoff, 2012). The dominant strand of empirical research has framed the fiscal question in terms of aggregate growth, following after Reinhart and Rogoff (2010) and Reinhart and Rogoff (2011), this literature has examined whether public debt above some threshold suppresses GDP growth.

Figure 2: Long-Run Trends in Debt



Note: Chained GDP-weighted cross-country means, anchored to the final-year cross-section to remove sample-entry effects. Sectoral debt from the IMF Global Debt Database. Weights are output-side real GDP (PWT 11.0).

The threshold hypothesis has proven fragile under scrutiny. Eberhardt and Presbitero (2015) and Chudik et al. (2017) demonstrate that the debt-growth relationship is characterized by substantial cross-country heterogeneity and that no common threshold exists across economies, even if country-specific tipping points cannot be ruled out. On this point, the empirical relationship between public debt and economic growth in the United States, western offshoot economies, and the OECD countries has also been exhaustively in the past decades (for example Dwyer Jr. (1982), Ahking and Miller (1985), Protopapadakis and Siegel (1987), King and Plosser (1985), Woo and Kumar (2015)). These empirical studies provide conflicting results, where some researchers find no significant relation between public debt and growth, while others find some relationship in their analysis with a limited data sample. One possible contributing factor is the paucity of readily available data. Another limitation is the lack of research for developing countries other than the U.S., OECD countries, and their western offshoots. In other words, an under looked factor is that countries selected for empirical studies in the prior decades are often well-developed countries with strong financial markets and access to foreign financial instruments. Such an arrangement naturally leads to selection bias, limiting the conclusions and inferences that can be drawn from such studies.

The distributional question has attracted far less systematic empirical attention. Azzimonti, de Francisco and Quadrini (2014) show that rising income inequality and financial integration jointly incentivize governments to accumulate debt, suggesting the two trends are endogenously linked rather than causally ordered. More recently, Grancini (2025) show in a heterogeneous-agent framework that high levels of public debt erode fiscal multipliers primarily through a factor-price channel: elevated real interest rates shift wealth toward households with lower marginal propensities to consume, weakening aggregate demand transmission. On the firm side, Islam and Nguyen (2024) argue that the crowding-out costs of public debt fall disproportionately on small and domestically owned firms, deepening the unequal playing field within the private sector.

These findings share a common implication that motivates this paper: the distributional consequences of public debt are mediated by institutional context, the composition of debt, and the composition of fiscal spending. A country that borrows to finance civilian public investment is likely to generate different distributional outcomes than one that borrows to service existing obligations or fund military expenditure, even at identical debt-to-GDP ratios. Similarly, the consequences of public debt expansion will differ markedly between a country with monetary and fiscal sovereignty operating under a flexible exchange rate and one constrained by fixed rates or external denomination of its obligations. Aggregate debt levels, taken alone, are theoretically insufficient to determine distributional outcomes.

This paper tests that proposition directly. Using a balanced panel of 109 countries over the period 1978–2019, we estimate the long-run relationship between public debt and three measures of income inequality, namely the top 10% income share, the top 1% income share, and the Gini coefficient, employing country-specific ARDL bounds testing with random-effects meta-analytic pooling, alongside a static panel fixed-effects specification as a complementary baseline. Our results yield null or insignificant effects of public debt on inequality in both the static and the long-run specification, a finding we treat as substantively informative rather than as a failure to detect an effect: debt accumulation in aggregate does not systematically worsen distributional outcomes across a broad cross-country panel. Income group analysis finds no group in which the debt-inequality relationship is concentrated, in either specification, and the one static coefficient that reaches significance, for the Gini in low-income countries, does not survive in the corresponding long-run estimate. Fiscal composition analysis, distinguishing civilian government spending from military expenditure, does not restore a debt effect that the aggregate specification fails to find; it does surface an income-group pattern in the Gini coefficient's long-run multiplier, but on a small and unstable subsample that we treat as a candidate for further scrutiny rather than as an established result. In place of a concentrated or composition-driven channel, this paper documents that neither is detected in this sample: the level of public debt, taken alone, is not a systematic determinant of income concentration.

2 Distributional Implications of Debt

2.1 Theoretical Framework

An array of research explain debt as the outcome of a political or economic struggle between different groups in the population who want to gain more control over resources. The reason debt is accumulated is that the group that is in power today may not be in power tomorrow, and debt is a way to take advantage of this temporary power. For instance, Song, Storesletten and Zilibotti (2012) argue that debt is a tool used to redistribute resources across generations. Alesina and Tabellini (1990) argue that debt represents a way to tie the hands of future governments that will have different preferences from the current one. Tabellini (1991) also illustrates how debt and social security differ as distributional instruments in an overlapping generations environment. In this vein, Bouton, Lizzeri and Persico (2020) develop a political economic model focusing on the relationship between entitlements and debt. They propose that both debt and entitlements are mechanisms through which temporarily dominant groups can extract resources from groups that will hold power in the future. Specifically, debt reallocates resources across time periods, while entitlements directly influence the future allocation of resources. Debrun, Jarmuzek and Shabunina (2020) suggest that the presence of endogenous entitlements reduces the incentive for politically powerful groups to accumulate debt, but this leads to an increase in overall government obligations. Furthermore, they argue that fiscal rules can have unintended consequences; if entitlements are not constrained and there are capital market frictions, imposing debt limits can increase total government obligations and result in worse outcomes for all groups.

A very different approach to understanding public debt is explored by Azzimonti, de Francisco and Quadrini (2014). They propose a multi-country model with incomplete markets, and they show that governments may choose higher public debt when financial systems are more integrated. Azzimonti, de Francisco and Quadrini (2014) argue that the incentives of governments to borrow rise both as financial markets become internationally integrated and as income inequality increases if this is associated with higher income risk.

Others argue that public spending financed by government deficits has reduced the inequality gap between low and high-income individuals in their respective countries (see Li and Filer, 2007; Van Bon, 2022). The central argument of this group of research is that "institutional quality" is crucial, and that advanced economies with rule-based governance can effectively utilize public spending to generate net positive outcomes. Countries that have successfully employed fiscal spending have seen it act as a catalyst for social transfers and resource reallocation through economic development (Ortiz-Ospina and Rose, 2016). High public debt, however, can also be seen as a cause of debt crises and political instability. Politicians in high-debt countries may prioritize balancing national accounts due to political pressure from voters, leading to a decrease in public spending policies and a subsequent increase in economic inequality above a certain debt-to-GDP threshold (Reinhart and Rogoff, 2011).

Most existing literature has predominantly examined either the factors influencing debt levels or the effects of overall economic growth on income inequality at the national or global level. However, the lack of studies addressing how debt impacts income inequality limits effective policy making, given the scarcity of knowledge on this direct relationship.

2.2 Debt Sustainability and Implications

The uncertainty surrounding fiscal policy and the ambiguous contours of a "safe" fiscal space have proven difficult to model, due in large part to the evolving nature of the agents and institutions involved. As Debrun et al. (2019) emphasize, debt sustainability remains an open frontier in macroeconomic analysis, one shaped not only by financial indicators but also by political dynamics. Indeed, sustainability is as much a political question as it is an economic one.

A central assumption in conventional macroeconomic frameworks is that sustainability is equivalent to solvency; that is, the state's ability and willingness to meet its future obligations. In this framework, countries can continue to borrow as long as the real interest rate on debt, r, is less than the growth rate of the economy, g. This condition has formed the basis for a new wave of thinking, most prominently advocated by Blanchard (2019, 2023) and Furman and Summers (2020), which suggests that public debt may entail little or no fiscal cost under prevailing conditions. If governments can roll over their debt at low interest rates without raising taxes, the debt-to-GDP ratio may decline over time even in the presence of deficits.

The empirical literature on the broader social and economic effects of public debt, however, remains contested. Much of the debate centers on whether there exists a specific debt-to-GDP threshold beyond which debt becomes destabilizing. While many studies implicitly or explicitly adopt such a threshold framework, others challenge this view. For instance, Acalin and Ball (2023) contest the standard postwar narrative that U.S. debt was reduced primarily through economic growth. Instead, they highlight the roles of interest rate pegs, surprise inflation, and tax increases in lowering the debt burden. In their view, the decline in U.S. public debt following World War II was less a function of benign growth dynamics than of active policy interventions.

On the question of a safe debt level, Debrun, Jarmuzek and Shabunina (2020) suggest that the United States remains within a "safe zone," even under the possibility of negative shocks. Their estimates indicate that debt levels as high as 160 percent of GDP may be sustainable, so long as real interest rates remain below growth rates and the maturity structure of debt is favorable. Mian, Straub and Sufi (2024) similarly argue that, prior to the COVID-19 pandemic, the U.S. could afford primary deficits above 2 percent of GDP and maintain debt levels over 120 percent of GDP. This view aligns with the broader emerging consensus that advanced economies with deep capital markets and monetary sovereignty face a more flexible debt constraint than previously believed.

Nonetheless, stabilizing the debt-to-GDP ratio remains a complex endeavor that requires deliberate policy action. The literature highlights several interlocking challenges: how to reconcile debt stabilization with political constraints, how to maintain growth while ensuring the intergenerational fairness of fiscal policy, and how to manage the distributional consequences of tax and spending decisions.

On one side of the debate, studies such as Boushey, Nunn and Shambaugh (2019), Ramey (2021), Fieldhouse and Mertens (2023), and Dynan and Elmendorf (2024) emphasize the growth-enhancing effects of public infrastructure investment, particularly when interest rates are low. On the other, researchers such as You and Dutt (1996), Chetty et al. (2017), Case and Deaton (2023), and Piketty and Saez (2019) foreground the distributive implications of fiscal policy, arguing that the composition of spending and taxation plays a pivotal role in shaping both inequality and long-run growth.

As macroeconomic conditions evolve and policy preferences shift, so too do the frameworks researchers and policymakers use to understand public debt. The literature remains rich and contested, reflecting not only competing economic theories but also the normative judgments that inevitably underlie debates over who pays, who benefits, and what counts as sustainable.

3 Related Literature

The renowned paper of Sargent and Wallace (1981) discusses the "monetary dominance" and "fiscal dominance" regimes said to emerge between the relationship of fiscal deficits and inflation. Given that the budget deficit is jointly determined by bond sales to the public and seigniorage created by a monetary authority if the monetary authority implements a monetary policy independently, then the fiscal authority faces a budget constraint imposed by the monetary authority when it formulates the fiscal policy. Under this circumstance, the monetary authority can control the money supply, and fiscal deficits do not lead to inflation. In contrast, in a fiscal dominance regime, the monetary authority cannot control the money supply, and fiscal deficits lead to inflation under such fiscal dominance.

In a widely cited paper, Reinhart and Rogoff (2010) provided long historical data series for the analysis of public-debt-to-GDP ratios and economic growth. Their finding that public-debt-to-GDP ratios above 90% are associated with markedly lower economic growth rates sparked a major debate. While several leading policymakers in the US and Europe directly referred to Reinhart and Rogoff (2010) in calling for immediate fiscal consolidation measures to reign in public debt, several groups of researchers used the data provided by Reinhart and Rogoff (2010) as well as newly constructed datasets to conduct extensive econometric tests on the impact of public debt levels on economic growth (e.g. Herndon, Ash and Pollin (2014); Woo and Kumar (2015)). Several papers argue that there is evidence for a negative causal effect of higher public-debt-to-GDP ratios on economic growth (e.g. Woo and Kumar (2015); Chudik et al. (2017)), and for a (close to) 90% threshold in the public-debt-to-GDP-ratio beyond which growth falls significantly (e.g. Checherita-Westphal and Rother (2012); Baum, Checherita-Westphal and Rother (2013)). While other studies acknowledge the stylized fact of a negative association between initial public debt levels and subsequent growth, they argue that the evidence for a causal effect running from higher public-debt-to-GDP to economic growth is weak at best (e.g. Panizza and Presbitero (2014)). Furthermore, several authors point to systematic differences in the (non-linear) impact of public debt on growth across countries, implying a lack of evidence for universal thresholds in the public-debt-to-GDP ratio beyond which growth falters (e.g. Eberhardt and Presbitero (2015); Eberhardt (2019); Bentour (2021)).

Likewise, the many empirical studies on the relationship between fiscal deficits and inflation provide contradictory and inconsistent results. For the United States, Hamburger and Zwick (1981) examine the deficit–money relationship from the period of 1954–1976, and conclude that budget deficits are inflationary. The relationship becomes stronger in the "Keynesian period" of 1961–1974. Dwyer Jr. (1982) uses quarterly data covering the 1953–1978 period to test the relationship between debt, price, and money; his results indicate that expected government deficits have no significance for future inflation. Ahking and Miller (1985) examine quarterly data from the period of 1947–1980, and present that the deficit–inflation relationship of the United States does exist during some specific periods. King and Plosser (1985) investigate the deficit–seigniorage relationship in terms of neoclassical macroeconomic models. They find little connection between fiscal deficits and seigniorage in the 1953–1982 period in the United States; they also estimate the deficit–seigniorage connection of 12 other industrial and developing countries, but still fail to demonstrate that the relationship is significant. Protopapadakis and Siegel (1987) examine the debt–money and the debt–inflation connections for 10 major advanced countries during the period of 1952–1987, and note that the association between debt growth and inflation is very weak. Haan and Zellhorst (1990), investigating 17 developing countries from 1961 to 1985, find no evidence to support the "fiscal dominance hypothesis," discovering that deficits are correlated to inflation during acute inflation periods.

Catão and Terrones (2005) model inflation as non-linearly related to fiscal deficits through the inflation tax base and estimate this relationship as intrinsically dynamic, using panel techniques that explicitly distinguish between short- and long-run effects of fiscal deficits. Their results, with a panel spanning 107 countries between 1960 and 2001, give a strong association between deficits and inflation among high-inflation and developing country groups, but not among low-inflation advanced economies.

In comparison, the relationship between inequality and the level of debt has been studied less. Much of the focus has been on the interplay between household debt and inequality (see for instance, Kumhof, Rancière and Winant (2015); Iacoviello (2008); Mason and Jayadev (2014)).

4 Data

The dataset is constructed from six primary sources: the Penn World Tables, the IMF Global Debt Database, the IMF World Economic Outlook, the World Inequality Database, the Standardized World Income Inequality Database, and the World Bank's World Development Indicators. These sources are merged at the country-year level; the estimation sample, determined by the coverage requirements described below, comprises 109 countries observed over a fixed 1978–2019 window. Country income classifications follow the World Bank Atlas method, assigned using each country's historical modal classification to avoid reclassification artifacts from single-year designations (World Bank, 2024*a*). The countries span low, middle, and high income groups across all major regions, providing a broad empirical basis for the analysis. Table 3 presents the distribution of countries by income group.[2]

[2]In calculating gross national income (GNI) per capita in U.S. dollars, the World Bank uses the Atlas method. The Atlas conversion factor for a given year is the average of a country's exchange rate for that year and the two preceding years, adjusted for the difference between the rate of inflation in the country and international inflation. GNI in U.S. dollars (Atlas method) for some year t is estimated by dividing the country's GNI in current prices (local currency) by the Atlas factor. The resulting GNI in U.S. dollars can then be divided by a country's midyear population to derive GNI per capita. For 2019, these definitions classify GNI per capita in U.S. dollars of less than 1,035 as "Low Income," 1,036 to 4,045 as "Lower Middle Income," 4,046 to 12,535 as "Upper Middle Income," and GNI per capita greater than 12,535 as "High Income."

Because country classifications change over time as GNI per capita evolves, using a single-year designation would introduce discontinuities into the sample stratification: a country that crossed a threshold in a single year would be assigned to different income groups depending on which year was chosen as the reference, with no substantive change in its structural characteristics. To avoid this, we assign each country its modal classification across all years for which it appears in the historical file. The modal classification represents the income group in which a country spent the largest number of years over the entire 1987–2024 period, and therefore reflects its dominant structural position rather than its transitory income status in any particular year. Where a country is tied across two groups, the higher group is assigned to avoid under-classifying countries that have experienced sustained development over the sample. The resulting distribution across the 137 countries in the estimation sample is reported in Table 3.

This approach is preferred to using the most recent classification for two reasons. First, the estimation sample spans several decades, over which many countries have graduated across thresholds; applying the end-of-sample classification would retrospectively impose a later income category on observations from earlier periods when the country had structurally different characteristics. Second, the research question concerns whether the debt-inequality relationship varies by level of development as a structural feature of the economy, not as a function of where a country happened to sit in a given year relative to a threshold.

4.1 Real Output and Macroeconomic Aggregates

Real GDP and related aggregates are drawn from the Penn World Tables (PWT) version 11.0 (Feenstra, Inklaar and Timmer, 2015). The primary measure of economic size used in weighting procedures is output-side real GDP (the PWT variable rgdpo), which values production at a country's own final demand prices and is the appropriate concept for cross-country comparisons of productive capacity. Real growth is computed from this series as the annual proportional change, and GDP per capita as output-side real GDP per person. Where PWT observations are missing, the growth and GDP per capita composites are completed from the IMF World Economic Outlook (WEO) (International Monetary Fund, 2023): the growth composite is supplemented with WEO real GDP growth, and the GDP per capita composite with the WEO series in purchasing power parity terms. The underlying PWT and WEO series are retained separately in the dataset, so the mixture of sources in each composite can be reconstructed by comparing the composite to its components even though no separate source indicator is carried into the final analysis file.

Trade openness is measured as the sum of the absolute export and import shares of GDP from the PWT national accounts data. The government consumption share of GDP provides the basis for the civilian spending share used in the fiscal decomposition analysis, constructed by subtracting the military expenditure share from the government consumption share. Government expenditure as a share of GDP, used elsewhere as a control for the overall size of the fiscal state, is measured from the IMF World Economic Outlook general government expenditure series.

4.2 Public Debt

The primary source for public debt is the IMF Global Debt Database (GDD) (Mbaye, Moreno-Badia and Chae, 2018), which provides sectoral debt series for up to 190 countries from 1950, covering general government, central government, household, non-financial corporate, and total private debt, each expressed as a percentage of GDP. The public debt variable is a composite constructed under a source hierarchy that prioritizes institutional breadth: general government debt from the GDD is used where available, central government debt from the GDD where general government coverage is missing, and gross general government debt from the WEO where neither GDD measure is reported. As with the output composites, a source indicator records the origin of each country-year observation. The hierarchy follows Reinhart and Rogoff (2011) in accepting central government debt as a proxy where general government data are incomplete, a substitution that primarily affects developing economies, while exploiting the broader coverage available in the GDD. Household, non-financial corporate, and total private debt are taken directly from the GDD and enter the descriptive analysis of debt composition. The WEO additionally supplies supplementary series for consumer price inflation and GDP in purchasing power parity terms.

4.3 Income Inequality

4.3.1 Top Income Shares

The primary inequality measures are top income shares from the World Inequality Database (World Inequality Database, 2024). We use the share of pre-tax national income accruing to the top 10% and the top 1% of the national pre-tax income distribution. These series are constructed under the Distributional National Accounts (DINA) framework (Blanchet, Chancel and Gethin, 2022), which combines tax records, household surveys, and national accounts to produce distributional estimates that are anchored to macroeconomic aggregates. This anchoring is a crucial methodological advantage: because WID estimates are constrained to sum to national income as measured in the System of National Accounts, they are substantially more comparable across countries and over time than survey-based estimates alone, which are subject to top-coding, non-response at the upper tail, and varying survey methodologies (Atkinson and Piketty, 2007). The DINA approach has been applied systematically across more than 100 countries in the World Inequality Report series (Chancel et al., 2022). For China, WID reports separate series for rural and urban populations; we average these to a single national observation for each year.

Top income shares are theoretically preferred to the Gini coefficient for this paper's research question for two reasons. First, they are more sensitive to changes at the upper tail of the distribution, where the fiscal and financial channels through which public debt operates are most likely to manifest (Atkinson, Piketty and Saez, 2011; Piketty and Saez, 2003). Second, the measurement error in top income shares is better characterized and more tractable than in survey-based Gini estimates: the primary limitation is the degree to which tax records capture top incomes, a concern that is extensively documented and partially addressed in the WID methodology (Alvaredo et al., 2013).

4.3.2 Gini Coefficient

The Gini coefficient is drawn from the Standardized World Income Inequality Database, version 9.91 (Solt, 2020). SWIID provides harmonized estimates of post-tax, post-transfer disposable income inequality for up to 199 countries from 1960 to the present. The harmonization procedure uses the Luxembourg Income Study (LIS) as the comparability benchmark, exploiting its consistent income definition and processing methodology across countries. Observations from other sources (Eurostat, the OECD Income Distribution Database, the World Bank's PovcalNet, the Socio-Economic Database for Latin America and the Caribbean, and national statistical offices) are then expressed relative to their proximate LIS observations using a predictive model that accounts for differences in income definition, unit of observation, and equivalence scale (Solt, 2020). This approach maximizes comparability without discarding the coverage gains available from non-LIS sources.

Because the harmonization involves imputation across surveys and years, SWIID reports 100 multiple imputations per country-year. We derive a point estimate as the mean Gini coefficient across all 100 imputations and its standard deviation as a measure of measurement uncertainty. Formally, combining inferences across all 100 imputations following Rubin's rules (Rubin, 1987) would be the statistically correct treatment; we adopt the simpler averaging procedure for computational tractability and note that the resulting standard errors may modestly understate true uncertainty, particularly in low-income countries where SWIID coverage relies most heavily on imputation across sparse survey years. The Gini coefficient serves as the secondary inequality measure and is used principally in robustness checks; the binding constraint on sample coverage is the WID, which provides wider and more historically continuous country coverage over the relevant sample period.

4.4 Control Variables

The choice of control variables follows the theoretical channels identified in the debt and inequality literatures, and is disciplined by the degrees-of-freedom constraints of the CS-DL and CS-ARDL estimators.

GDP per capita is included to control for the stage of economic development. The relationship between development and inequality is theoretically ambiguous. The Kuznets hypothesis predicts an inverted-U relationship, while the skill-biased technical change literature predicts monotone increases in the returns to education (Katz and Murphy, 1992; Goldin and Katz, 2008), but controlling for the level of income is necessary to avoid conflating debt effects with development-stage effects.

Government expenditure as a share of GDP controls for the size of the fiscal state, independent of how it is financed. The direction of the expenditure effect on inequality depends on the composition of spending: social transfers and public health and education expenditure are associated with reduced inequality (Dabla-Norris et al., 2015), while military and general administration spending are not. This motivates the fiscal decomposition specification, in which civilian government spending (the government consumption share less the military expenditure share) is distinguished from the aggregate expenditure share. Military expenditure as a share of GDP is sourced from the World Bank's World

Development Indicators (World Bank, 2024*b*) and used to construct the civilian spending share.

Life expectancy at birth is included as a broader measure of human development. It captures variation in public health infrastructure, nutrition, and living standards that is not fully absorbed by GDP per capita or government expenditure, and has been shown to be associated with income distribution through both human capital accumulation and labor market participation channels (Londono and Szekely, 2000). Life expectancy is preferred to the PWT human capital index for this purpose because the latter is constructed from assumed Mincer returns to schooling and therefore embeds a theoretical structure that may confound reduced-form estimates of the debt-inequality relationship.

Trade openness, constructed from the PWT national accounts shares as described above, is included to control for the distributional consequences of external sector integration. The effects of trade on inequality are theoretically contested: standard Heckscher-Ohlin reasoning predicts that openness reduces inequality in labor-abundant developing countries, while the empirical literature finds mixed results that depend strongly on the skill composition of the labor force and the nature of trading relationships (Goldberg and Pavcnik, 2007). Excluding trade openness would risk attributing to public debt any distributional effects that operate through the external sector.

Inflation is included because it is a distributional instrument in its own right. The redistributive incidence of inflation is asymmetric across the wealth distribution: it erodes the real value of nominal assets and liabilities and typically falls more heavily on lower-income households whose wealth is concentrated in liquid deposits rather than real assets (Doepke and Schneider, 2006). The composite inflation series draws on the World Development Indicators GDP deflator inflation rate, supplemented by WEO consumer price inflation where the former is missing. All control variables are sourced from the World Bank's World Development Indicators (World Bank, 2024*b*) unless otherwise noted.

4.5 Sample Construction

The estimation sample is defined over a common window applied to all countries rather than over country-specific spans. Candidate windows of between 30 and 50 years, beginning in every year from 1950 to 2024, were evaluated against the coverage criterion described below, and a window running from 1978 to 2019, providing 42 annual observations per country, was selected as offering a substantially larger qualifying sample than shorter or differently dated alternatives.

Within this window, a country is retained if four variables, the top 10% income share, the top 1% income share, the public debt-to-GDP ratio, and real GDP growth, each have no gap of more than two consecutive missing years and no more than 20 percent of observations missing overall (at most eight of the 42 years). Applying this criterion to the merged data yields 109 countries that satisfy it and form the estimation sample: 31 low-income, 26 lower-middle-income, 21 upper-middle-income, and 31 high-income countries, whose distribution is reported in Table 3.

Remaining interior gaps in a defined set of response and control variables, comprising public debt, growth, GDP per capita, government expenditure, the two income shares, the mean Gini coefficient, trade openness, composite inflation, WEO consumer price inflation, military expenditure, life expectancy, and the PWT government consumption share, are filled by Kalman smoothing under a local-level state-space model, fit separately to each country's own series. This is applied only where the series being filled itself has no gap longer than four consecutive years and no more than 20 percent of its observations missing; series failing this secondary, variable-specific threshold retain their remaining gaps rather than being smoothed. Because the imputed value at any point depends on the estimated state of the whole series rather than only on the most recently observed value, this procedure can introduce interpolated movement into a gap, unlike a simple carry-forward rule, and the resulting series should be read with that in mind where imputed spans are long relative to the surrounding observed data.

Because country classifications change over time as GNI per capita evolves, using a single-year designation would introduce discontinuities into the sample stratification: a country that crossed a threshold in a single year would be assigned to different income groups depending on which year was chosen as the reference, with no substantive change in its structural characteristics. To avoid this, we assign each country its modal classification across all years for which a classification is recorded in the underlying historical file, so that the assignment reflects a country's dominant structural position rather than its transitory income status in any particular year. The resulting distribution across the 109 countries in the estimation sample is reported in Table 3.

Table 3: Income Classification of Countries

Low Income	**Lower Middle Income**	**Upper Middle Income**	**High Income**
Bangladesh	Algeria	Argentina	Australia
Benin	Belize	Botswana	Austria
Burkina Faso	Bolivia	Brazil	Bahamas
Burundi	Cameroon	Chile	Bahrain
Central African Republic	Colombia	Costa Rica	Barbados
Chad	Congo	Dominica	Belgium
DR Congo	Cote d'Ivoire	Gabon	Canada
Equatorial Guinea	Egypt	Grenada	Cyprus
Ethiopia	El Salvador	Lebanon	Denmark
Gambia	Fiji	Malaysia	Finland
Ghana	Guatemala	Mauritius	France
Haiti	Guyana	Mexico	Germany
India	Honduras	Oman	Greece
Kenya	Indonesia	Panama	Iceland
Lao PDR	Iran	Saint Vincent and the Grenadines	Ireland
Madagascar	Jamaica	Seychelles	Italy
Malawi	Jordan	South Africa	Japan
Mali	Lesotho	Trinidad and Tobago	Luxembourg
Mauritania	Maldives	Turkey	Malta
Nepal	Morocco	Uruguay	Netherlands
Niger	Paraguay	Venezuela	New Zealand
Nigeria	Philippines		Norway
Pakistan	Sri Lanka		Portugal
Rwanda	Swaziland		Singapore
Sao Tome and Principe	Thailand		South Korea
Senegal	Tunisia		Spain
Sierra Leone			Sweden
Tanzania			Switzerland
Togo			United Arab Emirates
Zambia			United Kingdom
Zimbabwe			United States

Note: Income classifications follow the World Bank Atlas method. Each country is assigned its modal classification across all years for which a classification is recorded in the World Bank Analytical Classifications History file (World Bank, 2024*a*). The sample comprises 109 countries: 31 low income, 26 lower middle income, 21 upper middle income, and 31 high income.

5 Methods & Results

5.1 Debt and Inequality

General government debt levels exhibit deep heterogeneity across countries, wealthy and poor, alike. A way of illustrating this is presented in Figure 1, where we plot general government against against World Bank income classifications defined by Table 3. the distribution of public debt-to-GDP ratios (measuring each country's total government liabilities relative to the size of its economy) exhibits wide internal dispersion within each income bracket. Among low-income countries, debt ratios are generally moderate, although a few outliers suggest emerging fiscal pressures. Lower-middle-income countries display more spread, reflecting diverse macroeconomic conditions: some are undergoing fiscal consolidation, while others may be expanding public borrowing for development purposes or facing debt distress. Upper-middle-income countries, in contrast, show a surprising degree of clustering around relatively lower debt levels, possibly indicative of debt ceilings, external lending constraints, or recent policy reforms.

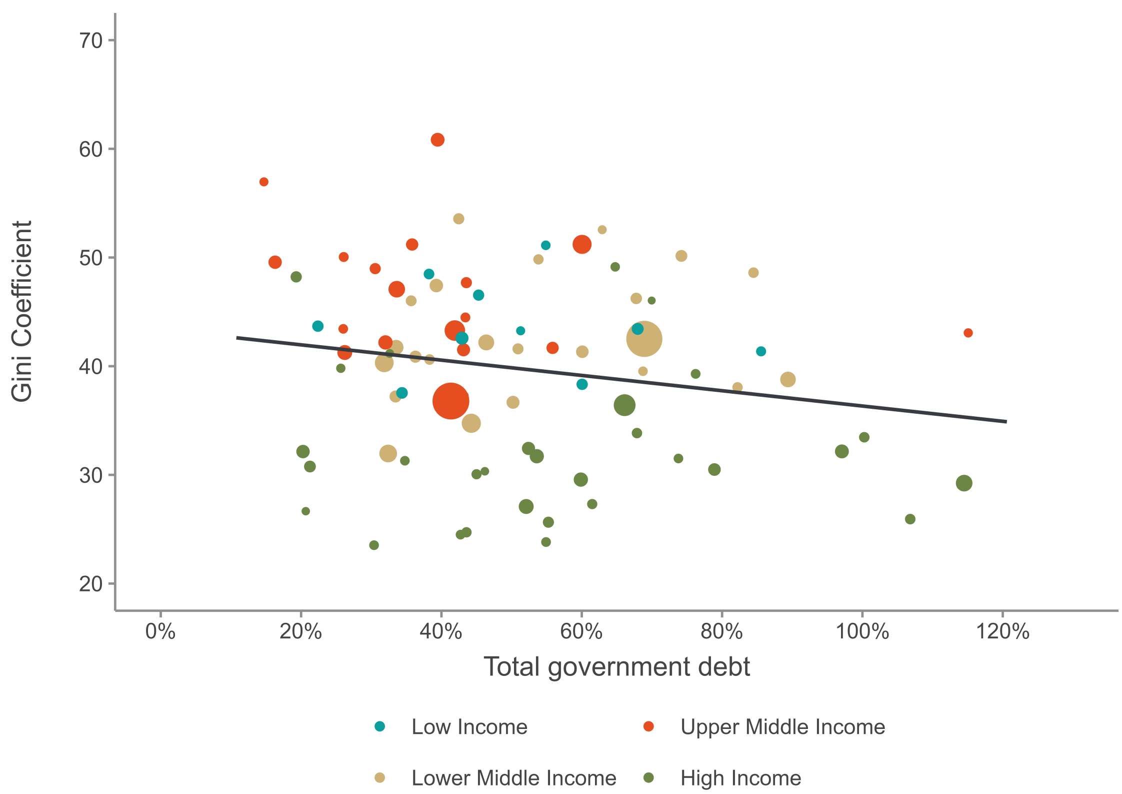
Figure 3: Gross Government Debt by IMF Income Classification

Note: Each dot represents a country. Size indicates population. Dotted line is the global average debt ratio; single lines represent income group level debt ratio average. Gross government debt and GDP attained from the IMF WEO from 1980-2019.

High-income countries present the most striking variation. While some maintain conservative debt profiles, others (including those with advanced welfare states or recent histories of fiscal stimulus) exceed 100 percent of GDP. The wide range among wealthier nations underscores that fiscal capacity and borrowing are not constrained solely by income level but by a complex interplay of historical, institutional, and geopolitical factors.

Marker size in the figure is proportional to each country's nominal GDP, allowing for a visual differentiation between economically larger and smaller states within each group. This scaling emphasizes the importance of considering both debt levels and economic size when evaluating fiscal sustainability. Crucially, the figure does not imply that higher income groups uniformly face greater debt burdens, but rather that fiscal heterogeneity is a feature of all income categories, one that defies simplistic generalizations and invites closer examination of national policy regimes, structural constraints, and the political economy of debt accumulation.

Figure 4: Gross Government Debt and Inequality



Note: Each dot represents a country. Size indicates population. Dotted line is the global average debt ratio; single lines represent income group level debt ratio average. Gross government debt and GDP attained from the IMF WEO from 1980-2019.

5.2 Panel Unit Root Properties

The CS-DL and CS-ARDL estimators, and the underlying ARDL bounds testing procedure of Pesaran and Shin (1999) and Pesaran, Shin and Smith (2001), require that each variable entering the long-run relationship be integrated of order zero or one. A variable integrated of order two or higher invalidates the bounds test critical values and undermines the interpretation of the long-run multiplier. Given the pronounced cross-country heterogeneity documented throughout this paper, the order of integration is assessed separately for each country rather than imposed uniformly across the panel.

For each country and each variable, we apply the Augmented Dickey-Fuller (ADF) test and the Phillips-Perron (PP) test for a unit root against the alternative of stationarity, first on the series in levels and, where the level test fails to reject, on its first difference. A series is classified $I(0)$ only where both tests reject the unit root null in levels at the 5 percent level; this conservative requirement of agreement guards against classifying a series as stationary on the strength of a single test. Failing that, a series is classified $I(1)$ where either test rejects the null in first differences, a more permissive criterion adopted because a series that remains nonstationary in differences by the agreement of both tests is a substantially stronger candidate for the $I(2)$ classification than one where the tests merely disagree. All remaining series are classified $I(2)$ or higher. The test is applied to a balanced sample of 109 countries, each contributing 42 annual observations across the two income share measures, public debt, and growth.

Table 4: Country-Level Order of Integration, by Variable

Variable	$I(0)$	$I(1)$	$I(2)+$
Top 10% income share	0	73	36
Top 1% income share	1	71	37
Public debt	0	95	14
GDP growth	26	83	0

Note: Counts of countries (of 109) by classified order of integration, based on country-by-country ADF and PP tests applied to levels and first differences at the 5 percent level.

Table 4 shows that GDP growth is the best-behaved of the four series. Roughly a quarter of countries (26 of 109, 23.9 percent) classify growth as stationary in levels, and the remaining three quarters achieve stationarity after first differencing, with no country requiring a higher order of integration. Public debt is almost uniformly $I(1)$: 95 of 109 countries (87.2 percent) fall into this category, against 14 countries (12.8 percent) where the series remains nonstationary even after differencing. Not a single country classifies public debt as $I(0)$, consistent with the persistence typically documented in debt-to-GDP ratios.

The two income share measures depart from this pattern. No country classifies the top 10 percent share as stationary in levels, and only one classifies the top 1 percent share this way. More consequentially, both series show a substantially higher incidence of $I(2)$ or higher classification than public debt or growth: 36 countries (33.0 percent) for the top 10 percent share and 37 countries (33.9 percent) for the top 1 percent share. These are not, for the most part, independent failures. Of the 40 countries where at least one income share measure classifies as $I(2)$ or higher, 33 (82.5 percent) show this result for both measures simultaneously, while 7 (17.5 percent) show it for a single measure alone. This concentration is consistent with the two series sharing a common source of persistence at the top of the income distribution, though the short country-specific samples, 42 annual observations each, also limit the power of univariate unit root tests to distinguish a genuine $I(2)$ process from an $I(1)$ process with strong low-frequency persistence. This caveat motivates the cross-sectionally augmented panel unit root test introduced below.

Restricting attention to the four series entering the country-specific ARDL specification (the two income share measures, public debt, and growth), 55 of 109 countries (50.5 percent) have all four series classified $I(0)$ or $I(1)$ and are therefore valid candidates for country-specific ARDL bounds estimation. The remaining 54 countries (49.5 percent) are excluded on the grounds of at least one $I(2)$ or higher series. These exclusions separate cleanly by source: 40 countries (74.1 percent of exclusions) are excluded because of the income share measures, and the remaining 14 (25.9 percent) because of public debt alone. No country in the sample is excluded on both grounds simultaneously, so nonstationary income shares and nonstationary public debt behave as distinct threats to identification rather than compounding within the same countries. Growth is never the source of exclusion, since no country classifies this series as $I(2)$ or higher.

The rate of ARDL validity also varies by income classification. High income countries have the highest share of valid country-specific specifications (20 of 31, 64.5 percent), and low income countries have the lowest (10 of 31, 32.3 percent). The two middle income groups fall in between, at close to half in each case (11 of 21, 52.4 percent, for upper middle income countries, and 14 of 26, 53.8 percent, for lower middle income countries). This gradient echoes a theme that recurs elsewhere in this paper: whether the underlying series are even well-behaved enough to support a bounds test differs systematically with development level, rather than following a single pattern common to all countries.

Table 5: Summary Statistics by Income Group

Income Group	Variable	Observations	Mean	SD	Min	Max
Upper Middle Income	Gini (imputed)	893	46.19	6.71	26.98	63.39
Upper Middle Income	Government Debt	893	0.41	0.28	0.04	2.38
Upper Middle Income	GDP	889	0.04	0.04	-0.16	0.21
Upper Middle Income	Population	893	103M	264M	657K	1.41B
High Income	Gini (imputed)	1410	32.12	7.02	20.52	51.53
High Income	Government Debt	1410	0.56	0.36	0.04	2.36
High Income	GDP	1392	0.03	0.03	-0.13	0.24
High Income	Population	1410	30.1M	53.3M	245K	328M
Low Income	Gini (imputed)	470	43.86	4.13	34.76	52.16
Low Income	Government Debt	470	0.50	0.31	0.03	1.69
Low Income	GDP	460	0.04	0.06	-0.50	0.35
Low Income	Population	470	10.2M	7.54M	577K	42.9M
Lower Middle Income	Gini (imputed)	1081	42.66	5.86	29.20	54.71
Lower Middle Income	Government Debt	1081	0.52	0.32	0.02	2.61
Lower Middle Income	GDP	1077	0.04	0.04	-0.22	0.26
Lower Middle Income	Population	1081	77.6M	205M	1.14M	1.38B
Total	Gini (imputed)	3854	39.77	8.69	20.52	63.39
Total	Government Debt	3854	0.50	0.33	0.02	2.61
Total	GDP	3818	0.04	0.04	-0.50	0.35
Total	Population	3854	57.9M	173M	245K	1.41B

Note: Gini coefficients are from the Standardized World Income Inequality Database (SWIID) v9.3. Government debt and GDP figures are based on IMF World Economic Outlook estimates. Population data are sourced from the World Bank. Observations represent country-year pairs.

5.3 Estimation

We first estimate the relationship between public debt and income concentration using a static panel specification with country and year fixed effects. For country i in year t,

$$y_{it} = \alpha_i + \lambda_t + \gamma\, d_{it} + \boldsymbol{\beta}'\mathbf{x}_{it} + u_{it}, \tag{5.1}$$

where y_{it} is one of three measures of income concentration, d_{it} is the ratio of public debt to GDP, and $\mathbf{x}_{it}$ is a vector of controls. The country effects absorb time-invariant determinants of the level of concentration and the year effects absorb common shocks. We cluster standard errors by country. Public debt enters as a ratio to GDP and the income shares as fractions of national income, so γ is the percentage point change in the share associated with a one percentage point change in debt-to-GDP. Gini coefficients are in index points.

Estimates are reported in Table 6 for three measures of concentration and five specifications. Specification (1) controls for growth, life expectancy, and the interaction of debt with growth. Specification (2) adds government consumption and inflation. Specification (3) replaces government consumption with its military and civilian components and replaces the headline inflation rate with a composite inflation index. Specification (4) adds the interaction of debt with civilian spending, and specification (5) adds trade openness and the investment share.

Table 6: Public Debt and Income Concentration

	Top 10% share	Top 1% share	Gini
(1) Baseline	0.0060	0.0032	0.693
	(0.0052)	(0.0040)	(0.444)
(2) Fiscal size	0.0082	0.0036	0.786*
	(0.0055)	(0.0045)	(0.417)
(3) Mil. vs civ.	0.0077	0.0037	0.711
	(0.0063)	(0.0044)	(0.440)
(4) Debt × civ.	0.0200**	0.0142*	0.457
	(0.0097)	(0.0084)	(0.635)
(5) Full	0.0180*	0.0125	0.653
	(0.0092)	(0.0082)	(0.647)

Note: Coefficient on public debt from (5.1), with standard errors clustered by country in parentheses. All columns include country and year fixed effects. Gini coefficients are in index points. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

The coefficient on public debt is positive in every specification for all three measures of concentration, ranging from 0.0060 to 0.0200 for the top 10 percent share, 0.0032 to 0.0142 for the top 1 percent share, and 0.457 to 0.786 for the Gini. It is not statistically distinguishable from zero in specifications (1) through (3) for any of the three measures. Significance emerges only once the interaction between debt and civilian spending enters the specification. In specification (4), the coefficient is 0.0200 ($p = 0.041$), significant at the 5 percent level, for the top 10 percent share, and 0.0142 ($p = 0.094$), significant at the 10 percent level, for the top 1 percent share; the Gini coefficient in this specification, 0.457 ($p = 0.473$), remains far from significance. In the full specification, the coefficient for the top 10 percent share, 0.0180 ($p = 0.053$), falls just short of the 5 percent threshold, while the coefficients for the top 1 percent share, 0.0125 ($p = 0.131$), and the Gini, 0.653 ($p = 0.315$), are not significant. Of the 15 estimates in Table 6, one is statistically significant at the 5 percent level and four at the 10 percent level. In the full specification, a ten percentage point increase in the ratio of public debt to GDP is associated with an increase of 0.18 percentage points in the top 10 percent share, significant at the 10 percent level, and an increase of 0.12 percentage points in the top 1 percent share, which does not reach conventional significance.

The interaction between debt and growth is not statistically significant in any of the 15 estimates, with p-values ranging from 0.146 to 0.687. In the full specification, military expenditure enters at 0.190 ($p = 0.242$) for the top 10 percent share and 0.217 ($p = 0.204$) for the top 1 percent share, and the interaction between debt and civilian spending at -0.063 ($p = 0.117$) and -0.054 ($p = 0.233$). Composite inflation is significant for the top 10 percent share at the 10 percent level ($p = 0.064$) and for the Gini at the 5 percent level ($p = 0.011$).

Table 7 reports specification (3) estimated separately within each income group. Of the twelve coefficients on public debt, one is statistically significant at the 5 percent level: 1.403 for the Gini in low-income countries ($p = 0.043$). The coefficient for lower-middle-income countries is negative across all three measures of concentration (-0.0078 for the top 10 percent share, -0.0109 for the top 1 percent share, and -0.677 for the Gini), though not significantly so in any of them. The coefficient for upper-middle-income countries is positive across all three measures but likewise falls short of conventional significance, including for the top 10 percent share (0.0029, $p = 0.720$). We find no evidence that the relationship between debt and concentration concentrates in the income groups with the least fiscal space.

Table 7: Public Debt and Income Concentration by Income Group

	Top 10% share	Top 1% share	Gini
High income	0.0098	-0.0006	-0.422
	(0.0125)	(0.0044)	(1.013)
Upper middle income	0.0029	0.0050	0.896
	(0.0081)	(0.0083)	(0.868)
Lower middle income	-0.0078	-0.0109	-0.677
	(0.0079)	(0.0090)	(0.872)
Low income	0.0028	0.0011	1.403**
	(0.0093)	(0.0065)	(0.661)

Note: Coefficient on public debt from specification (3), estimated separately within each income group. Standard errors clustered by country in parentheses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

The income group estimates in Table 7 do not support the prediction that the relationship concentrates where fiscal space is most constrained. The two middle-income groups carry coefficients of opposite sign, positive for upper-middle-income countries and negative for lower-middle-income countries, with neither approaching significance. The one significant coefficient, for the Gini in low-income countries, is not accompanied by a corresponding relationship for either income share in the same income group, nor by a significant Gini coefficient in any other income group.

Table 8 reports estimates of (5.1) in first differences, which removes the country effects and any confounder that changes slowly within countries. The coefficient on the change in public debt is -0.00063 ($p = 0.694$) for the top 10 percent share, 0.00046 ($p = 0.758$) for the top 1 percent share, and -0.041 ($p = 0.703$) for the Gini. No regressor is significant for either income share; for the Gini, only the change in inflation enters significantly, at 0.012 ($p = 0.021$). The level estimates in Table 6 therefore derive from variation that annual first differencing removes.

Table 8: First-Difference Estimates

	Δ Top 10%	Δ Top 1%	Δ Gini
Δ Public debt	-0.00063	0.00046	-0.041
	(0.00160)	(0.00149)	(0.108)
Δ Growth	0.00134	-0.00050	-0.016
	(0.00137)	(0.00090)	(0.021)
Δ Life expectancy	0.00013	0.00002	0.008
	(0.00016)	(0.00016)	(0.005)
Δ Military exp.	-0.02007	0.01540	0.645
	(0.03935)	(0.03232)	(1.097)
Δ Civilian spend.	-0.00742	-0.00350	-0.065
	(0.00556)	(0.00504)	(0.270)
Δ Inflation	0.00000	0.00001	0.012^{**}
	(0.00001)	(0.00001)	(0.005)

Note: All columns include year fixed effects with standard errors clustered by country. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

The estimates in Table 6 do not identify a channel. The coefficient on public debt reaches significance, at conventional but modest levels, only in the specifications that include the interaction between debt and civilian spending, and only for the top 10 percent and top 1 percent shares; it is never significant for the Gini. The terms through which the composition argument of section 2 would operate are not themselves significant: military expenditure is positive but insignificant throughout, the interaction between debt and civilian spending is negative but insignificant throughout, and the interaction between debt and growth is insignificant throughout. The specifications that include the debt-civilian spending interaction produce a public debt coefficient two to three times larger than the specifications that do not, while the interaction term itself, and the spending components that motivate it, carry no independent explanatory weight.

The first-difference estimates in Table 8 give no evidence of a relationship at annual frequency. Since first differencing removes the country effects and any determinant of concentration that changes slowly within countries, the level estimates are identified from low-frequency within-country variation that the differenced specification does not use. This does not invalidate the level estimates: a relationship operating over a longer horizon than one year would not appear in first differences. It does mean that the static specification cannot distinguish a long-run relationship from a persistent confounder, and that we cannot recover an adjustment horizon from it.

Two features of the specification limit what (5.1) can establish. It constrains γ to be common across the countries within each column, which the dispersion in Table 7 gives reason to doubt. And it relates concentration to debt contemporaneously, so it recovers no long-run parameter if either series is persistent. Section 5.4 estimates the relationship separately for each country and separates short-run dynamics from the long-run relation.

5.4 Country-Specific ARDL Estimation

The fixed-effects estimates of Table 6 and Table 7 constrain the coefficient on public debt to be common across countries and impose a static relation between debt and inequality. This section relaxes both restrictions and estimates the long-run relation country by country, separately for the top decile share, the top percentile share, and the Gini coefficient. The exercise yields a bound rather than an effect. For the top decile share, the point estimate of the pooled long-run multiplier implies a small decline in concentration rather than the rise observed over the sample, and the upper end of the 95 percent interval attributes at most 5.5 percent of the observed rise to debt accumulation. The same conclusion, indistinguishable from zero at conventional levels, holds for the top percentile share and for the Gini coefficient.

5.4.1 Specification

For each country i we estimate an autoregressive distributed lag model in levels, following Pesaran and Shin (1999) and Pesaran, Shin and Smith (2001):

$$y_{it} = c_i + \sum_{j=1}^{p_i} \phi_{ij}\, y_{i,t-j} + \sum_{\ell=1}^{4} \sum_{j=0}^{q_{i\ell}} \beta_{i\ell j}\, x_{i\ell,t-j} + \varepsilon_{it}, \tag{5.2}$$

where y_{it} is, in turn, the top decile share, the top percentile share, and the Gini coefficient, and the four regressors are the public debt-to-GDP ratio, real GDP growth, life expectancy at birth, and the product of debt and growth. The model is estimated separately for each of the three outcomes, so that the lag structure, the cointegration evidence, and the long-run multiplier are all country- and outcome-specific rather than imposed jointly across measures of concentration.

Each term in (5.2) absorbs a distinct source of variation. The country intercept c_i absorbs the time-invariant institutional and structural determinants of the level of concentration, which the descriptive evidence of Table 2 shows to differ by more than ten percentage points across income groups. The own lags ϕ_{ij} absorb the persistence of the income distribution, which is high enough that a specification omitting them attributes to the regressors variation that is accounted for by the previous year's outcome. Growth and life expectancy absorb the development-stage channels set out in section 4, so that the debt coefficient is not identified from the covariation of debt with the level of income or with human development. The interaction term permits the effect of debt to depend on the growth environment in which it is accumulated.

Lag orders $(p_i, q_{i1}, \ldots, q_{i4})$ are selected by the Akaike Information Criterion over a maximum order of three in each argument. The long-run multiplier on regressor ℓ is

$$\theta_{i\ell} = \frac{\sum_{j=0}^{q_{i\ell}} \beta_{i\ell j}}{1 - \sum_{j=1}^{p_i} \phi_{ij}}, \tag{5.3}$$

with standard errors obtained by the delta method from the estimated coefficient covariance matrix. The denominator of (5.3) is the negative of the speed of adjustment in the associated conditional error correction representation, $\lambda_i = \sum_j \phi_{ij} - 1$, so the long-run multiplier and the adjustment dynamics are two readings of the same estimated autoregressive polynomial rather than independent pieces of evidence. The debt ratio is measured in percentage points of GDP and the two income shares in percentage points of national income, so θ_{i1} reads as the percentage point change in the share, or, for the Gini coefficient, the index-point change, associated with a permanent one percentage point change in the ratio of public debt to GDP. We report multipliers throughout alongside the movement they imply over the debt expansion recorded in Table 1, where public debt rises 61.7 percentage points of GDP between the pre-1979 and recent periods while the top decile share rises 8.4 percentage points.

Of the countries satisfying the order-of-integration requirement of subsection 5.2, 69 enter the exercise. All 69 are estimated for the top decile and top percentile shares. Nine are excluded from the Gini specification for having too few annual observations once the requisite lags are accounted for, leaving 60. The estimation samples are long relative to the requirements of country-specific time series work: every country contributes 42 annual observations to the two income share specifications, and a median of 42 (minimum 25) to the Gini specification. The selected dynamics are parsimonious for the income shares and considerably less so for the Gini coefficient. For the top decile share, 54 countries select a single own lag, 12 select two, and 3 select three; 20 of the single-lag models are the fully static specification $(1, 0, 0, 0, 0)$, with no lagged terms on any regressor. For the top percentile share the corresponding counts are 48, 17, and 4, of which 23 are static. For the Gini coefficient, 12 countries select a single own lag, 40 select two, and 8 select three, and only 1 is static.

5.4.2 Evidence of Long-Run Relations

The long-run multiplier in (5.3) has an equilibrium interpretation only where the variables are cointegrated. The bounds F test evaluates the joint nullity of the levels terms in the conditional error correction form against the alternative of a long-run relation, computed for every country under an unrestricted intercept and no trend. Table 9 reports the outcomes for each of the three measures of concentration.

The null of no long-run relation is rejected less often than not for all three measures. At the 5 percent level, rejection occurs in 17 of 68 countries with a valid statistic for the

Table 9: ARDL Bounds Test Results, by Measure of Concentration

Conclusion	**Top 10% share**	**Top 1% share**	**Gini**
Cointegration, $p \leq 0.01$	9	14	9
Cointegration, $p \leq 0.05$	8	8	12
Weak evidence, $0.05 < p \leq 0.10$	8	8	7
No evidence, $p > 0.10$	43	38	29
Test could not be evaluated	1	1	3
Total estimated	69	69	60

Note: Bounds F tests following Pesaran, Shin and Smith (2001) with $k = 4$ regressors, computed under an unrestricted intercept and no trend. The bounds test could not be evaluated at the selected lag order for the United Arab Emirates (top decile share), Burkina Faso (top percentile share), and Brazil, Singapore, and Thailand (Gini). These countries retain long-run multipliers in Table 11 but carry no cointegration verdict.

top decile share (25.0 percent), 22 of 68 for the top percentile share (32.4 percent), and 21 of 57 for the Gini coefficient (36.8 percent); at the 10 percent level the corresponding counts are 25 (36.8 percent), 30 (44.1 percent), and 28 (49.1 percent). The test statistics are centered well below the customary critical values: means of 3.17, 3.76, and 3.59 for the top decile share, the top percentile share, and the Gini coefficient respectively, with medians of 2.71, 3.33, and 3.31, and ranges running from 0.43 to 10.70, from 1.05 to 10.11, and from 0.77 to 7.96.

Rejection is unrelated to development level for any of the three outcomes. Median statistics for the top decile share run from 1.79 in lower-middle-income countries to 3.69 in low-income countries, with high-income and upper-middle-income countries in between at 3.54 and 2.30. For the top percentile share the lowest median falls in high-income countries, at 3.06, against 4.46 in low-income countries. For the Gini coefficient the lowest median falls in upper-middle-income countries, at 2.66, against 4.72 in lower-middle-income countries. No income group is consistently the most or the least likely to show evidence of a long-run relation across the three outcomes, and the ordering itself is different for each one.

The nine countries rejecting most strongly for the top decile share are drawn from three of the four income groups and divide close to evenly by sign: the long-run multiplier on public debt is negative in five and positive in four. The implied movement over the sample debt expansion ranges from a decline of 9.3 percentage points in the United Kingdom to a rise of 18.9 points in Bangladesh, a span that argues against reading any single country's estimate as representative of a general pattern.

5.4.3 Long-Run Multipliers

None of the three pooled estimates is distinguishable from zero. Applied to the sample debt expansion, the estimate for the top decile share implies a decline of 0.74 percentage points, against an observed rise of 8.4 points: the point estimate points in the opposite direction from the observed increase in concentration, and the upper end of the 95 percent interval, $+0.46$ points, accounts for at most 5.5 percent of it. Testing the pooled estimate against

Table 10: Countries Rejecting No Cointegration at the 1 Percent Level, Top Decile Share

Income Group	**Country**	F	$\hat{\theta}_{\mathbf{debt}}$	**Implied pp**
High Income	United States	10.70	-0.034	-2.09
High Income	Finland	8.25	0.253	15.63
High Income	United Kingdom	8.10	-0.151	-9.34
Low Income	Bangladesh	6.65	0.307	18.93
High Income	Italy	6.28	0.049	3.05
High Income	Netherlands	6.00	0.004	0.28
High Income	Denmark	5.59	0.045	2.76
High Income	Austria	5.41	-0.022	-1.35
Upper Middle Income	Chile	5.18	-0.027	-1.69

Note: Long-run debt multipliers from (5.3). The implied column gives the percentage point movement in the top decile share over the 61.7 percentage point rise in public debt recorded in Table 1, against 8.4 percentage points observed. All nine countries have a stable autoregressive polynomial.

Table 11: Long-Run Multiplier on Public Debt: Distribution and Pooled Estimate

	Top 10% share	**Top 1% share**	**Gini**
k	68	68	52
Median	-0.0046	0.0005	-0.730
IQR	$[-0.051,\ 0.037]$	$[-0.022,\ 0.052]$	$[-4.65,\ 8.47]$
Pooled $\hat{\theta}$	-0.0119	0.0041	-0.529
SE	0.0099	0.0055	1.354
p	0.227	0.458	0.696
I^2 (%)	83.9	83.7	89.2
Implied pp	-0.74	0.25	-0.33

Note: Random-effects meta-analysis by restricted maximum likelihood with inverse-variance weights, restricted to countries with a stable autoregressive polynomial and a defined multiplier. The implied column converts the pooled estimate into the movement over the 61.7 percentage point rise in public debt recorded in Table 1. Heterogeneity: top decile share, $Q(67) = 269.5$, $p < 0.001$; top percentile share, $Q(67) = 215.7$, $p < 0.001$; Gini, $Q(51) = 426.7$, $p < 0.001$.

the hypothesis $\theta \geq 0.0204$, the multiplier that would be required for debt accumulation to account for 15 percent of the observed rise, returns $z = 3.27$ and $p < 0.001$; testing against $\theta \geq 0.0504$, corresponding to 37 percent of the observed rise, returns $p < 0.0001$. The data are informative about the absence of a large positive long-run effect on the top decile share rather than merely uninformative about its presence.

For the top percentile share, the same debt expansion implies a rise of 0.25 percentage points at the point estimate, with a 95 percent interval running from -0.41 to $+0.91$ points. Tested against the same two thresholds, a multiplier at or above 0.0204 is rejected with $z = 2.99$ ($p = 0.001$), and a multiplier at or above 0.0504 is rejected with $p < 0.0001$. For the Gini coefficient, the pooled estimate implies a decline of 0.33 points, with a 95 percent interval running from -3.18 to $+2.12$; on the scale of index points rather than fractional shares this interval is too wide to be read the same way, a point taken up further below.

Twelve of 68 country-specific multipliers on public debt are individually significant at the 5 percent level for the top decile share, and they divide seven negative to five positive. The significant negative cases are the United Arab Emirates (-0.716), the United Kingdom (-0.151), Luxembourg (-0.384), the Bahamas (-0.091), Zambia (-0.071), Malta (-0.062), and Cote d'Ivoire (-0.051); the significant positive cases are Bangladesh (0.307), Cyprus (0.088), Italy (0.049), Lebanon (0.047), and Jordan (0.026). For the top percentile share, nine of 68 are individually significant, three negative (Luxembourg, -0.628; the United Kingdom, -0.088; Malta, -0.042) and six positive (South Africa, 0.677; South Korea, 0.508; Bangladesh, 0.215; Bahrain, 0.078; Greece, 0.057; Jordan, 0.027). For the Gini coefficient, four of 52 stable multipliers are individually significant: Brazil (46.2) and Bolivia (27.9) positive, El Salvador (-22.9) and Japan (-7.4) negative, magnitudes that reflect the coefficient's units rather than an economically distinct scale of effect. The sign division in all three lists is close to what a null relation with heterogeneous precision would generate, and none is dominated by one sign or one income group.

The remaining regressors show a different pattern for the two income shares than for the Gini coefficient. Growth pools to a positive and significant long-run multiplier for both the top decile share (0.0293, SE 0.0145, $p = 0.044$) and the top percentile share (0.0203, SE 0.0084, $p = 0.016$), with no cross-country heterogeneity in either case ($I^2 = 0$). Life expectancy pools to a positive and highly significant multiplier for the top percentile share (0.00135, SE 0.00024, $p < 0.001$) but not for the top decile share ($p = 0.421$). The interaction between debt and growth is significant and negative for the top percentile share (-0.0306, SE 0.0110, $p = 0.005$) but not for the top decile share ($p = 0.570$). This bears on the conditional-effect interpretation advanced from the static estimates in **??**, where the marginal effect of growth on inequality was read as declining in the debt ratio: that pattern does not survive in the long-run estimate for the top decile share, but it does appear in the estimate for the top percentile share, and the two income share measures do not agree on this point. For the Gini coefficient, none of the three secondary regressors approaches significance, and the standard errors on growth and the debt-growth interaction are an order of magnitude larger than the corresponding estimates.

5.4.4 Development Level as the Leading Alternative

The mainstream account of the debt-inequality relation predicts that its severity increases as institutional capacity and fiscal space contract, so that the effect should concentrate in the lower-income groups. This prediction is directly testable as a moderator restriction on the country-specific multipliers. Table 12 reports a random-effects meta-regression of the public debt multiplier on World Bank income group, for each of the three outcomes.

Table 12: Long-Run Debt Multiplier by Income Group

Income Group	**Top 10% share**	**Top 1% share**	**Gini**
Low income	0.0209 [10]	0.0209 [10]	0.442 [5]
	(0.0280)	(0.0201)	(4.165)
Lower middle income	−0.0096 [16]	−0.0020 [15]	0.181 [11]
	(0.0192)	(0.0127)	(3.652)
Upper middle income	−0.0074 [12]	−0.0012 [13]	1.311 [13]
	(0.0248)	(0.0146)	(2.869)
High income	−0.0267 [30]	0.0109 [30]	−1.764 [23]
	(0.0163)	(0.0103)	(2.033)
Omnibus test of equality			
$Q_M(3)$, p, R^2	2.24, 0.523, 0%	1.41, 0.704, 0%	0.87, 0.833, 0%

Note: Group-specific pooled estimates from a random-effects meta-regression by restricted maximum likelihood without an intercept. Standard errors in parentheses; bracketed figures are the number of countries contributing to each cell. R^2 is the share of between-country heterogeneity in the multiplier accounted for by income group.

No group-specific estimate is significant for any of the three outcomes, and income group accounts for essentially none of the between-country heterogeneity in any of them. The largest point estimate in absolute value is the high-income coefficient on the top decile share, at -0.0267, and its sign runs against the direction predicted for the group with the greatest fiscal space; the corresponding high-income coefficient for the Gini, -1.764, points the same way.

This result does not corroborate the one significant income-group finding in the static specification. In Table 7 the debt coefficient for low-income countries on the Gini was 1.403, significant at the 5 percent level. Its long-run counterpart here is 0.442, less than a third the magnitude, with a standard error of 4.165 and $p = 0.916$. The static coefficient is identified from a pooled within-country association at annual frequency under a common-slope restriction; the long-run estimate comes from a country-specific dynamic model estimated on five low-income countries with a defined multiplier. The divergence is consistent with the reading developed throughout this section, that the static annual association and the country-specific long-run relation answer different questions, and that a result significant in the former need not survive in the latter.

Four features of the estimates admit readings other than the one advanced above, and the pooled estimate is recomputed under two of them in Table 13.

Unstable autoregressive polynomials. One country's autoregressive polynomial is estimated with $\sum \hat{\phi}_j \geq 1$ for the top decile share (South Africa, at 1.009) and one for the top percentile share (Algeria, at 1.034); such cases leave the denominator of (5.3) non-positive and the multiplier without an equilibrium interpretation. Eight countries fall in this category for the Gini coefficient, with polynomials as high as 1.769 (India) and

Table 13: Pooled Long-Run Debt Multiplier Under Alternative Restrictions

Subsample	**Top 10% share**	**Top 1% share**	**Gini**
All countries	-0.0119 [68] (0.0099)	0.0041 [68] (0.0055)	-0.529 [52] (1.354)
Cointegrated, $p \leq 0.10$	-0.0103 [25] (0.0180)	0.0397* [30] (0.0216)	-0.882 [23] (2.208)
Cointegrated, $p \leq 0.05$	-0.0058 [17] (0.0229)	0.0625* [22] (0.0327)	-0.103 [16] (3.570)

Note: Restricted maximum likelihood random-effects estimates with inverse-variance weights. Bracketed figures are k. $^*p < 0.10$ (two-sided).

implied multipliers running into the hundreds of index points (Egypt, 410.4; the United States, 566.6). Unlike an earlier draft of this analysis, these cases are excluded from every pooled estimate reported above as a matter of course rather than folded in and removed only as a robustness check. For the Gini coefficient this affects eight of 60 estimated countries, enough that the coefficient's basic numerical behavior, on a variable measured in index points rather than as a fractional share, is a limitation of (5.3) worth flagging on its own terms.

Selection on the outcome through the cointegration pretest. Table 13 restricts the pooled estimate to countries showing at least weak evidence of cointegration. For the top decile share the restriction moves the estimate from -0.0119 to -0.0103 at the 10 percent threshold and to -0.0058 at the 5 percent threshold, in both cases still centered near zero. For the top percentile share the restriction moves the estimate from 0.0041 to 0.0397 ($p = 0.066$) at the 10 percent threshold and to 0.0625 ($p = 0.056$) at the 5 percent threshold, in both cases falling just short of conventional significance. For the Gini coefficient the restricted estimates remain centered near zero throughout. The correlation between the bounds test statistic and the absolute value of the multiplier is modest for all three outcomes (0.13, 0.07, and 0.07 respectively), weaker than would raise serious concern about the pretest selecting on effect size, though the mean absolute multiplier among cointegrated countries is still more than double the mean among the remainder for both income share measures (0.26 against 0.11 for the top decile share, 0.49 against 0.12 for the top percentile share). The near-significant result for the top percentile share under the cointegrated restriction should accordingly be read with some caution, as resting on a subsample chosen partly on the magnitude of the effect it is used to isolate.

Estimation precision against genuine heterogeneity. The heterogeneity statistics reported in Table 11 measure dispersion in the estimated coefficients relative to their sampling error, not the share of variation in concentration that debt accounts for, and much of the apparent dispersion reflects imprecision. For the top decile share, 18 of 68 countries have confidence intervals lying entirely within ± 0.05 and 17 have intervals wider than ± 0.2, a span that over the sample debt expansion corresponds to more than 12 percentage points of the top decile share. For the top percentile share, 26 of 68 lie within ± 0.05 and 17 are wider than ± 0.2. The country-specific estimates individually cannot distinguish a null effect from a substantial one in a large share of cases, which is why the pooled estimate and the equivalence bound, rather than the distribution of country-level point estimates, carry the weight of the argument.

5.4.5 Adjustment Dynamics

The speed of adjustment λ_i gives the fraction of a deviation from the long-run relation closed within one year, with stability requiring $-1 < \lambda_i < 0$. Table 14 reports the distribution for each outcome.

Table 14: Speed of Adjustment, by Measure of Concentration

Range	Top 10% share	Top 1% share	Gini
$\lambda < -1$	0	0	0
$-1 \leq \lambda < -0.30$	29	44	7
$-0.30 \leq \lambda < -0.10$	29	13	20
$-0.10 \leq \lambda < 0$	10	11	25
$\lambda \geq 0$	1	1	8
Total	**69**	**69**	**60**

Note: $\lambda_i = \sum_j \hat{\phi}_{ij} - 1$ from the conditional error correction representation. The cases with $\lambda \geq 0$ are the unstable polynomials identified above.

Adjustment is rapid for both income share measures and considerably slower for the Gini coefficient. For the top decile share, 68 of 69 countries fall in the stable range, with a median coefficient of -0.237 and a median half-life of 2.6 years (interquartile range 1.4 to 4.6 years); group medians run from -0.139 in upper-middle-income countries to -0.288 in high-income and lower-middle-income countries. For the top percentile share, adjustment is faster still: a median coefficient of -0.385 and a median half-life of 1.4 years (interquartile range 0.9 to 3.0 years), with group medians from -0.248 in low-income countries to -0.468 in lower-middle-income countries. For the Gini coefficient, 52 of 60 countries fall in the stable range, with a median coefficient of -0.107 and a median half-life of 6.1 years (interquartile range 2.9 to 16.7 years); group medians run from -0.020 in lower-middle-income countries to -0.128 in high-income countries. The error correction models account for a median of 39 percent of the annual variance in the top decile share, 38 percent in the top percentile share, and 75 percent in the Gini coefficient, the last reflecting the coefficient's larger scale rather than a materially better-fitting model.

Adjustment toward the estimated long-run relation is therefore rapid for both measures of top income concentration, running against the reading, common in the fiscal literature and adopted in an earlier draft of this analysis, that the distributional consequences of public debt unfold over horizons too long to detect in samples of this length. On the evidence here the horizon is not the binding constraint for either income share measure. The long-run multiplier on public debt is indistinguishable from zero for all three outcomes, income group accounts for none of the cross-country dispersion in any of them, and for a majority of countries the variables do not stand in a long-run relation at all; where they do, adjustment is a matter of a few years rather than decades.

5.4.6 Robustness to Fiscal Composition

The specification in (5.2) omits the composition of government spending, the channel through which section 2 argues public debt could bear on distribution even where its aggregate level does not. A second specification adds military expenditure and non-military government consumption as regressors, entering with the same maximum lag order as the other variables. The additional regressors reduce the estimation sample: 60 of 69 countries are estimated for the top decile and top percentile shares (six skipped for insufficient observations, three yielding a rank-deficient fit with no defined long-run relation), and 43 of 69 for the Gini coefficient (25 skipped, one rank-deficient).

The pooled long-run multiplier on public debt is essentially unchanged. Under this specification it is 0.0009 for the top decile share ($k = 57$, $p = 0.925$), 0.0022 for the top percentile share ($k = 52$, $p = 0.709$), and -0.403 for the Gini coefficient ($k = 36$, $p = 0.731$), each closer to zero than its counterpart in Table 11 and none significant. Military expenditure and non-military spending are themselves imprecisely estimated for both income share measures: military expenditure pools to -0.107 ($p = 0.794$) and non-military spending to -0.101 ($p = 0.071$) for the top decile share, and to 0.044 ($p = 0.831$) and -0.025 ($p = 0.438$) respectively for the top percentile share. Neither spending component reaches the 5 percent level for either income share measure, and adding the composition of spending does not restore a debt effect that the baseline specification fails to find.

Cointegration is somewhat more common under this specification: the null of no long-run relation is rejected at the 5 percent level in 34.0 percent of countries for the top decile share, 44.6 percent for the top percentile share, and 40.0 percent for the Gini coefficient, each several points above the corresponding baseline rate in Table 9. The composition specification is also estimated on a smaller and different set of countries, so the two rates are not a controlled comparison of the same sample under two specifications.

One result under this specification warrants a caution rather than a claim. For the Gini coefficient, income group accounts for 43.2 percent of the between-country heterogeneity in the debt multiplier ($Q_M(3) = 19.2$, $p < 0.001$), driven by a lower-middle-income coefficient of -11.3 ($p = 0.003$) and an upper-middle-income coefficient of 6.1 ($p = 0.002$) of opposite sign. No corresponding pattern appears for either income share measure ($R^2 = 0$ percent in both). Given that the Gini multiplier is already the least numerically stable of the three outcomes, and that this result rests on small samples within each group (four lower-middle-income and eight upper-middle-income countries, several with standard errors an order of magnitude larger than their point estimates), this pattern is reported as a candidate for further scrutiny rather than as evidence of income-group heterogeneity in the composition channel.

The aggregate level of public debt is not a quantitatively important determinant of top income concentration in this sample, for either measure of concentration or for the Gini coefficient, and this conclusion is robust to the inclusion of the composition of government spending. The remainder of the paper turns to the composition of borrowing itself, and to the institutional setting in which fiscal expansion occurs.

5.5 Discussion

Taken together, the results complicate simple narratives that treat public debt as uniformly detrimental or uniformly benign. The relationship is conditional: it depends on the stage of development, institutional capacity, and the growth environment in which debt is accumulated. For advanced economies with well-developed financial markets, rising debt levels may coincide with higher inequality, potentially reflecting the regressive incidence of debt-financed fiscal consolidation, or the unequal distributional effects of accommodative monetary policies that accompany high debt ratios. For lower-middle-income countries, debt accumulation may exacerbate inequality through more direct channels, including increased exposure to external shocks and the need for austerity measures imposed by international creditors.

By contrast, the absence of a strong relationship in upper-middle-income countries suggests that institutional factors (such as targeted social transfers, fiscal rules, or debt ceilings) may mitigate the distributive consequences of public borrowing. The evidence that human capital investments reduce inequality in wealthier states but exacerbate disparities in poorer ones is particularly instructive. It points to the role of access: where educational and health resources are broadly available, public debt that finances these expenditures has equalizing effects; where access is restricted, debt-financed investments can reinforce existing divides.

Our findings also speak to the broader debate on fiscal space and debt sustainability. If the distributive consequences of debt vary so markedly across income groups, then one-size-fits-all policy prescriptions, such as universal debt thresholds, are misleading. Instead, the political economy context (monetary sovereignty, exchange rate regime, and institutional quality) determines whether public debt functions as a stabilizing or destabilizing force in the long run.

6 Conclusion

This paper has examined the complex interplay between public debt, economic growth, and income inequality across 82 countries over nearly five decades. Contrary to the notion of a universal debt threshold beyond which growth and equity outcomes deteriorate, our results highlight deep heterogeneity. Debt does not exert a uniform effect on inequality; instead, its distributive impact depends on structural conditions, the level of development, the interaction with economic growth, and the institutional environment in which fiscal policy operates.

The central finding of this study is that the transmission of debt into inequality is conditional on institutional quality. This leads us to advance an institutional-first approach to debt management: public debt should not be treated as a universal stabilization tool, but as an instrument that requires a specific level of administrative and institutional maturity to operate safely. The heterogeneity documented throughout our analysis is not a statistical nuisance; it is the result itself. Where institutions are robust (where tax administration is effective, budget processes are transparent, and central banks operate with credible independence) debt can serve as a macro-stabilizer during downturns, channeling resources toward social protection and productive investment. Where these foundations are absent, the same fiscal instrument risks amplifying inequality through elite capture, regressive consolidation, and external vulnerability.

Three specific conclusions follow. First, debt accumulation is most likely to exacerbate inequality in lower-middle-income and high-income economies, albeit through different mechanisms. In lower-middle-income countries, the "austerity channel" dominates: weak institutional capacity means that debt obligations translate into expenditure cuts that fall disproportionately on the poor, while external creditor pressures further constrain progressive fiscal responses. In high-income economies, the inequality-increasing effects of debt operate through more subtle channels, regressive fiscal adjustments and the unequal distributional consequences of accommodative monetary policies that accompany high debt ratios. For both groups, the implication is the same: institutional context determines whether debt is stabilizing or destabilizing.

Second, economic growth on its own is not necessarily equalizing. In low-income countries, growth often coincides with widening disparities, and our finding that human capital investments can exacerbate inequality in these settings points to processes of elite capture. Where access to education and health resources remains stratified, debt-financed investments reinforce existing divides rather than narrowing them. This underscores the need for institutional prerequisites (broad-based access to public services, progressive tax structures, and accountable governance) before debt can function as a vehicle for inclusive development.

Third, the exchange rate regime analysis reveals that monetary sovereignty creates fiscal *capacity* without determining fiscal *outcomes*. In the short run, sovereign countries with floating exchange rates actually exhibit stronger positive associations between debt and inequality. The long-run ARDL estimates hint at a reversal, with sovereign countries showing more inequality-reducing debt multipliers on average, but the evidence is not statistically decisive. The critical insight is that the absence of external discipline does not, by itself, translate into progressive fiscal choices. What matters is not whether a country *can* run deficits freely, but how it *chooses* to spend and that choice is shaped by institutional quality.

These findings yield a clear policy prescription, differentiated by institutional maturity. For low-income and lower-middle-income countries at an early stage of institutional development, the priority must be capacity building: strengthening tax administration, improving budget transparency, establishing credible and independent central banking, and broadening access to public services. Until these foundations are set, the austerity channel poses too great a risk, and debt should be deployed with extreme caution and directed toward the institutional investments that will expand future fiscal space. For upper-middle-income and advanced economies that have already achieved a degree of institutional maturity, the evidence supports a more confident use of debt as a countercyclical instrument provided that the composition of spending is designed to reach broad segments of the population rather than accruing disproportionately to asset holders and elites.

For international policy debates, the findings caution decisively against reliance on simple debt-to-GDP thresholds or cross-country averages. The 90 percent threshold popularized by Reinhart and Rogoff (2010), whatever its merits as a stylized fact, obscures the institutional heterogeneity that our analysis reveals to be the dominant feature of the debt-inequality landscape. Context-sensitive assessments of fiscal space, ones that account for institutional capacity, exchange rate arrangements, and the functional composition of public spending, are essential.

Future research should extend the analysis along two dimensions. First, incorporating more granular data on the functional composition of public spending would clarify which categories of expenditure drive the equalizing or disequalizing effects of debt across institutional contexts. Second, modeling the interaction between domestic political institutions and external financial constraints would illuminate the mechanisms through which institutional quality mediates the debt-inequality relationship. Doing so would further clarify the conditions under which public debt can support both growth and equity in the global economy, and would provide the empirical foundation for the institutional-first framework we advocate here.

7 References

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